THE THEOREM OF THE CUBE IDENTITY FOR A LINE BUNDLE ON AN ABELIAN VARIETY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Theorem more-morphisms-theorem-of-the-cube (canonical tag 0BF4). cited at proof lines 1915-1918. The cube theorem applied to M.

Source reference. Original theorem 0BF4: sources/stacks-chow/tags.txt, source lines 14372–14372. Cited at proof lines 1915-1918 as the theorem applied to the difference module M; load-bearing.

Lemma 0.2. *Let k be a field. Let A be an abelian variety over k. Let $\mathcal{L}$ be an invertible $\mathcal{O}_A$-module. Then there is an isomorphism*

$$m_{1,2,3}^*\mathcal{L} \otimes m_1^*\mathcal{L} \otimes m_2^*\mathcal{L} \otimes m_3^*\mathcal{L} \cong m_{1,2}^*\mathcal{L} \otimes m_{1,3}^*\mathcal{L} \otimes m_{2,3}^*\mathcal{L}$$

of invertible modules on $A \times_k A \times_k A$ where $m_{i_1,\ldots,i_t} : A \times_k A \times_k A \to A$ is the morphism $(x_1, x_2, x_3) \mapsto \sum x_{i_j}$.

Proof. Apply the theorem of the cube (More on Morphisms, Theorem 0BF4) to the difference

$$\mathcal{M} = m_{1,2,3}^*\mathcal{L} \otimes m_1^*\mathcal{L} \otimes m_2^*\mathcal{L} \otimes m_3^*\mathcal{L} \otimes m_{1,2}^*\mathcal{L}^{\otimes -1} \otimes m_{1,3}^*\mathcal{L}^{\otimes -1} \otimes m_{2,3}^*\mathcal{L}^{\otimes -1}$$

This works because the restriction of $\mathcal{M}$ to $A \times A \times e = A \times A$ is equal to

$$n_{1,2}^*\mathcal{L} \otimes n_1^*\mathcal{L} \otimes n_2^*\mathcal{L} \otimes n_{1,2}^*\mathcal{L}^{\otimes -1} \otimes n_1^*\mathcal{L}^{\otimes -1} \otimes n_2^*\mathcal{L}^{\otimes -1} \cong \mathcal{O}_{A \times_k A}$$

where $n_{i_1,\ldots,i_t} : A \times_k A \to A$ is the morphism $(x_1, x_2) \mapsto \sum x_{i_j}$. Similarly for $A \times e \times A$ and $e \times A \times A$. □

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